

MACHINE LEARNING

DECISION TREE

HYPER TUNING PARAMETER ANALYSIS

S.NO	Criterion	Splitter	max_features	R_Score value
1.	Default(squared_error)	Default(best)	Default(none)	0.89116
2.	Squared_error	Random	Sqrt	-0.87219
3.	Squared_error	Best	Sqrt	0.60021
4.	Squared_error	Random	Log2	0.86443
5.	Squared_error	Best	Log2	-0.06361
6.	Squared_error	Random	none	0.94927
7.	Friedman_mse	Best	Sqrt	0.28984
8.	Friedman_mse	Random	Sqrt	0.72644
9.	Friedman_mse	Best	Log2	0.44295
10.	Friedman_mse	Random	Log2	0.91886
11.	Friedman_mse	Best	None	0.91739
12.	Friedman_mse	Random	None	0.77746
13.	Absolute_error	Best	Sqrt	0.93682
14.	Absolute_error	Random	Sqrt	0.79516
15.	Absolute_error	Best	Log2	0.85657
16.	Absolute_error	Random	Log2	0.45573
17.	Absolute_error	Best	None	0.92653
18.	Absolute_error	Random	None	0.38265
19.	Poisson	Best	Sqrt	0.34682
20.	Poisson	Random	Sqrt	0.40800
21.	Poisson	Best	Log2	0.93773
22.	Poisson	Random	Log2	0.74620
23.	poisson	Best	None	0.93121
24.	poisson	Random	None	0.93585

CONCLUSION: Decision tree- hyper parameter
(criteria=squared_error,splitter=random,max_features=none) is a “**BEST MODEL**”

